

RethinkDB:

A NoSQL Database for Real-Time Applications

RethinkDB is an open source, scalable database that makes building real-time apps dramatically easier.

With the heterogeneous as well as unstructured data on the World Wide Web (WWW) increasing in leaps and bounds, traditional database engines are facing numerous issues related to schema management, concurrency, database integrity, security, parallel read-write operations, resource optimisation and many others. Real-time applications receive lots of traffic from different channels including social media, mail groups, satellites, the Internet of Things (IoT), etc. Such traffic is generally unstructured and voluminous, and cannot be handled by classical relational database management systems (RDBMS). To cope with such issues of Big Data, which include velocity, volume and variety (the 3Vs of Big Data), Not Only SQL (NoSQL) databases that could handle unstructured and heterogeneous data with higher performance came into existence.

Today, there are different types of NoSQL databases in diverse categories to help specific applications achieve a higher degree of performance and accuracy (Table 1).

RethinkDB as a document-oriented NoSQL database

RethinkDB (<https://www.rethinkdb.com>) is a high-performance NoSQL database for real-time applications with an effective Web based administrative panel and user interface. The database system provides excellent features for document storage based applications, and is termed as a document-oriented database. RethinkDB provides features for storage, retrieval and management of semi-structured data to achieve the minimum delay in database operations. The key concept and paradigm behind this document-oriented database is the storage and handling of documents in unstructured or semi-structured formats. All document formats can be encapsulated with the encoding of data in document-oriented NoSQL databases. The encoding of data includes YAML, BSON, XML and JSON, along with the other binary forms including PDF, MS Word, spreadsheets and many others.



Table 1: Types of NoSQL databases

Taxonomy	NoSQL database
Document store	ArangoDB, Couchbase, BaseX, Clusterpoint, CouchDB, DocumentDB, MarkLogic, IBM Domino, MongoDB, RethinkDB Qizx
key-value store (Classical)	ArangoDB, Keyspace, Flare, Aerospike, RAMCloud, SchemaFree, quasardb
key-value store (Ordered)	Actord, Lightcloud, FoundationDB, InfinityDB, LMDB, Luxio, NMDB, MemcacheDB, TokyoTyrant
Key-value cache	Memcached, Apache Ignite, eXtreme Scale, Hazelcast, Infinispan, JBoss Cache, Coherence, Repcached, Velocity
Tuple store	Coord, Apache River, GigaSpaces
Object database	Perst, Objectivity/DB, DB4O, Shoal, ZopeDB
Data structures	Redis
Wide column store	Bigtable, KAI, Amazon DynamoDB, HBase, Hypertable, Cassandra, Druid, OpenNeptune, KDI, Qbase
key-value store (Ordered)	Oracle NoSQL Database, DovetailDB, Riak, Dynomite, Dynamo, Voldemort, SubRecord